Digital Political Advertising During Live Sports Events

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1 Digital Political Advertising During Live Sports Events

1.0.1 Engagement & Advertising Performance Analysis

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Project Type: Digital Advertising Analytics
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1.1 Project Overview

This project analyzes digital political advertising performance during live sports events. The goal of the analysis is to evaluate how advertising engagement varies across sports leagues, digital platforms, audience demographics, and different moments during a game.

Live sports broadcasts provide a unique advertising environment because viewers are highly engaged and frequently interact with second-screen platforms such as mobile devices, social media, sports applications, and streaming services. This behavior creates opportunities for digital political campaigns to reach audiences through targeted advertising strategies.

By analyzing engagement metrics such as click-through rates, impressions, advertising recall, and audience interaction, this project identifies patterns in advertising effectiveness and highlights the most impactful environments for digital political advertising during live sports events.

1.2 Introduction

Live sports events represent one of the most concentrated and engaged audiences in modern media. Millions of viewers watch sporting events while simultaneously interacting with digital platforms such as social media, sports applications, streaming services, and betting platforms. This second-screen behavior creates a powerful opportunity for digital advertising strategies, including political messaging during major sporting events.

Research shows that a large percentage of sports fans actively use mobile devices while watching games. This behavior allows advertisers to reach audiences through multiple platforms at the same time, increasing exposure, engagement, and message recall.

The objective of this project is to analyze digital political advertising performance during live sports broadcasts. The analysis focuses on how engagement rates, click-through rates, and advertising recall vary across different sports leagues, advertising platforms, audience segments, and moments during games.

Using a structured dataset of sports advertising metrics, this notebook explores patterns in advertising performance and identifies which sports environments and digital platforms generate the strongest audience engagement. The findings can help inform strategic decisions about where and when political campaigns should invest in digital advertising during live sporting events.

1.3 Dataset Overview

The dataset used in this analysis contains advertising performance metrics collected across multiple sports leagues and digital platforms.

Each row in the dataset represents a specific advertising observation, including the sport, platform, game moment, geographic market, and target audience demographic.

Key variables include:

- **Sport:** The sports league or competition (NFL, NBA, MLB, NHL, College Football, Soccer).
- **Platform:** The digital platform where the advertisement appeared (social media, streaming services, sports apps, betting apps, etc.).
- **Game Moment:** The timing of the advertisement during the sporting event.
- **Mobile Usage Percentage:** Estimated percentage of viewers using a second screen during the event.
- **Ad Impressions:** Total number of advertisement views.
- **Click-Through Rate (CTR):** Percentage of viewers who clicked the advertisement.
- **Engagement Rate:** Overall audience interaction with the advertisement.
- **Ad Spend:** Estimated advertising investment.
- **Ad Recall:** Percentage of viewers who remembered the advertisement after exposure.

These variables allow us to explore how advertising performance varies across sports environments, platforms, and audience segments.

[20]:
```python
# 1. Import libraries
import pandas as pd
import matplotlib.pyplot as plt

# 2. Load dataset
df = pd.read_csv("sports_political_ads_dataset_large.csv")

# 3. Preview data
df.head()
```

[20]:
```
    Sport      Platform  Game_Moment State Audience_Age_Group  Mobile_Usage_%  \
0     NHL       YouTube    Post-Game    TX              35-54              80
1     NHL    Sports App  4th Quarter    NY              25-44              83
2     NBA  Connected TV  3rd Quarter    FL              18-34              76
3     NHL   Betting App     Pre-Game    TX              35-54              78
4  Soccer     Streaming  1st Quarter    PA              25-44              80
```

	Ad_Impressions_M	CTR_%	Engagement_Rate_%	Ad_Spend_M	Ad_Recall_%
0	252	2.23	5.70	14.18	11
1	563	1.69	6.27	8.67	18
2	208	2.12	5.33	24.48	9
3	284	1.59	5.46	23.98	18
4	535	2.77	8.08	7.44	19

[21]:
```
# 4. Check dataset structure
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 11 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Sport               250 non-null    object
 1   Platform            250 non-null    object
 2   Game_Moment         250 non-null    object
 3   State               250 non-null    object
 4   Audience_Age_Group  250 non-null    object
 5   Mobile_Usage_%      250 non-null    int64
 6   Ad_Impressions_M    250 non-null    int64
 7   CTR_%               250 non-null    float64
 8   Engagement_Rate_%   250 non-null    float64
 9   Ad_Spend_M          250 non-null    float64
 10  Ad_Recall_%         250 non-null    int64
dtypes: float64(3), int64(3), object(5)
memory usage: 21.6+ KB
```

[22]:
```
# 5. Summary statistics
df.describe()
```

[22]:

	Mobile_Usage_%	Ad_Impressions_M	CTR_%	Engagement_Rate_%	\
count	250.000000	250.000000	250.000000	250.000000	
mean	74.568000	368.596000	2.391160	8.355680	
std	8.671919	131.303692	0.623957	2.072686	
min	60.000000	151.000000	1.210000	5.060000	
25%	67.000000	256.500000	1.920000	6.590000	
50%	75.000000	366.000000	2.400000	8.315000	
75%	82.000000	489.000000	2.945000	10.120000	
max	89.000000	599.000000	3.500000	11.980000	

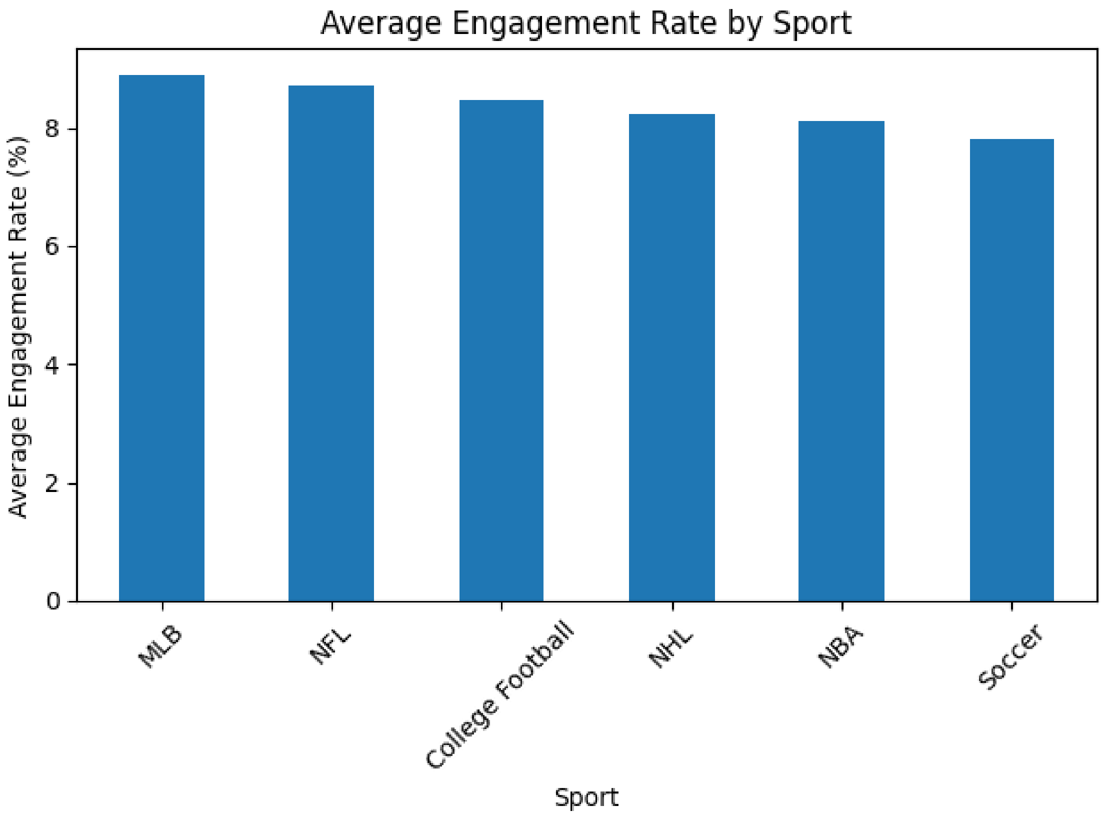
	Ad_Spend_M	Ad_Recall_%
count	250.000000	250.000000
mean	14.700440	12.860000
std	5.913711	3.831858
min	5.050000	7.000000
25%	9.450000	10.000000

```
50%      15.210000    13.000000
75%      19.730000    16.000000
max      24.920000    19.000000
```

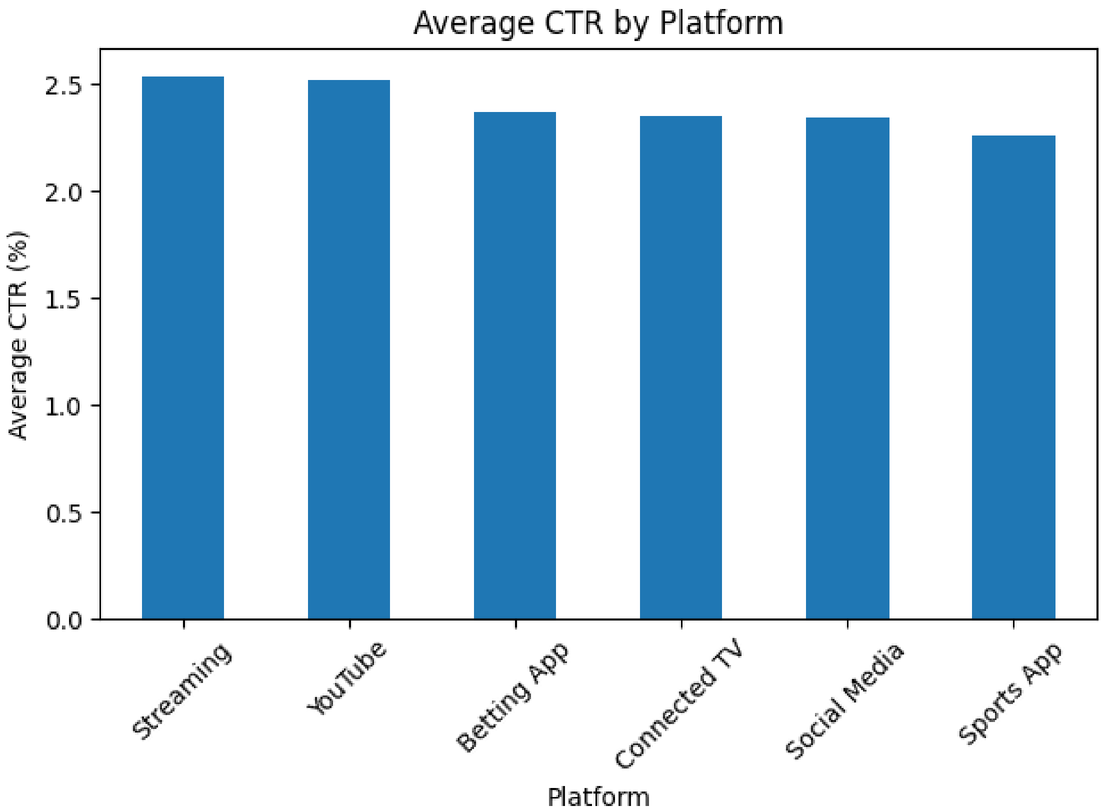
[23]:
```
# 6. Check for missing values
df.isnull().sum()
```

[23]:
```
Sport                  0
Platform               0
Game_Moment            0
State                  0
Audience_Age_Group     0
Mobile_Usage_%         0
Ad_Impressions_M       0
CTR_%                  0
Engagement_Rate_%      0
Ad_Spend_M             0
Ad_Recall_%            0
dtype: int64
```

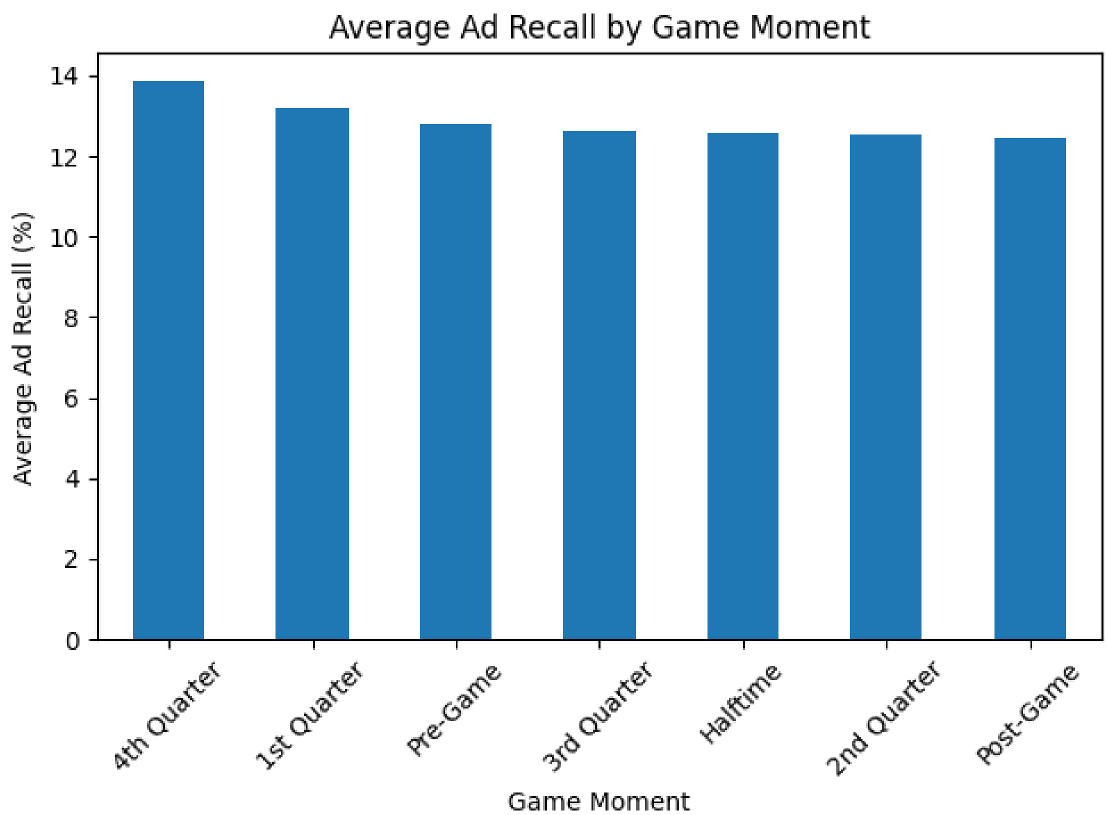
[24]:
```
# 7. Engagement Rate by Sport
sport_engagement = df.groupby("Sport")["Engagement_Rate_%"].mean().
 ↪sort_values(ascending=False)

plt.figure()
sport_engagement.plot(kind="bar")
plt.title("Average Engagement Rate by Sport")
plt.xlabel("Sport")
plt.ylabel("Average Engagement Rate (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



[25]:
```
# 8. CTR by Platform
platform_ctr = df.groupby("Platform")["CTR_%"].mean().
↪sort_values(ascending=False)

plt.figure()
platform_ctr.plot(kind="bar")
plt.title("Average CTR by Platform")
plt.xlabel("Platform")
plt.ylabel("Average CTR (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

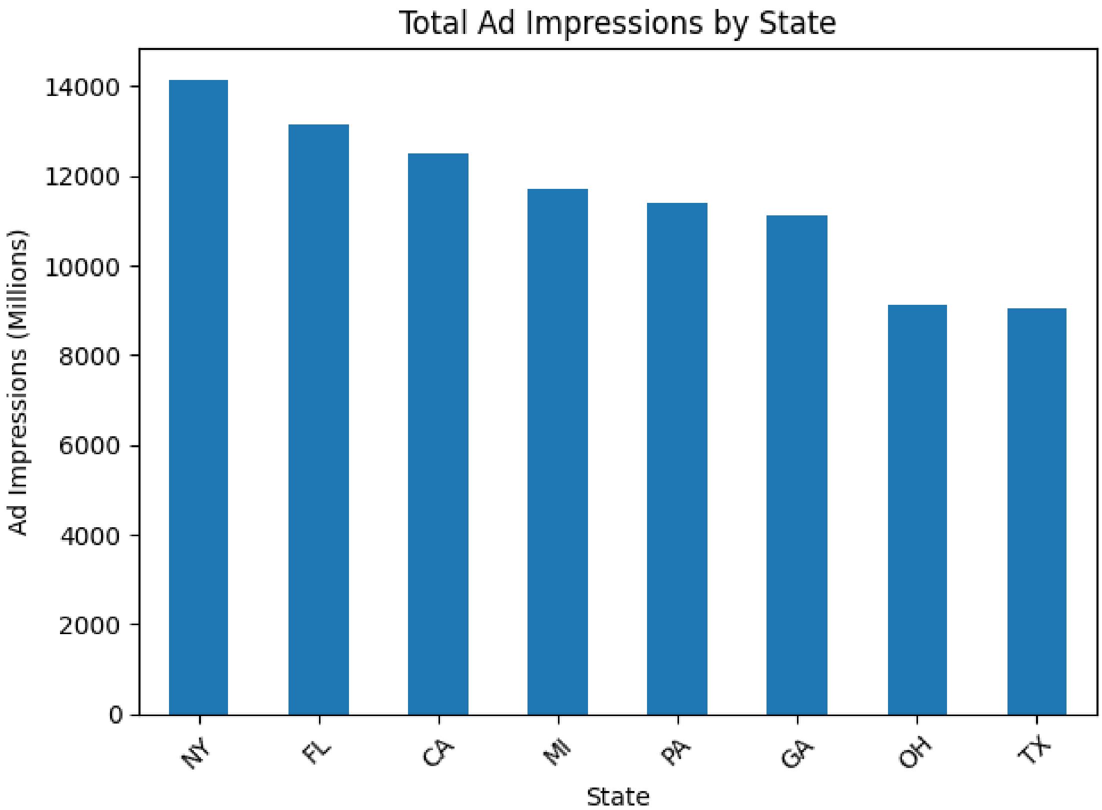


[26]:
```
# 9. Ad Recall by Game Moment
moment_recall = df.groupby("Game_Moment")["Ad_Recall_%"].mean().
 ↪sort_values(ascending=False)

plt.figure()
moment_recall.plot(kind="bar")
plt.title("Average Ad Recall by Game Moment")
plt.xlabel("Game Moment")
plt.ylabel("Average Ad Recall (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

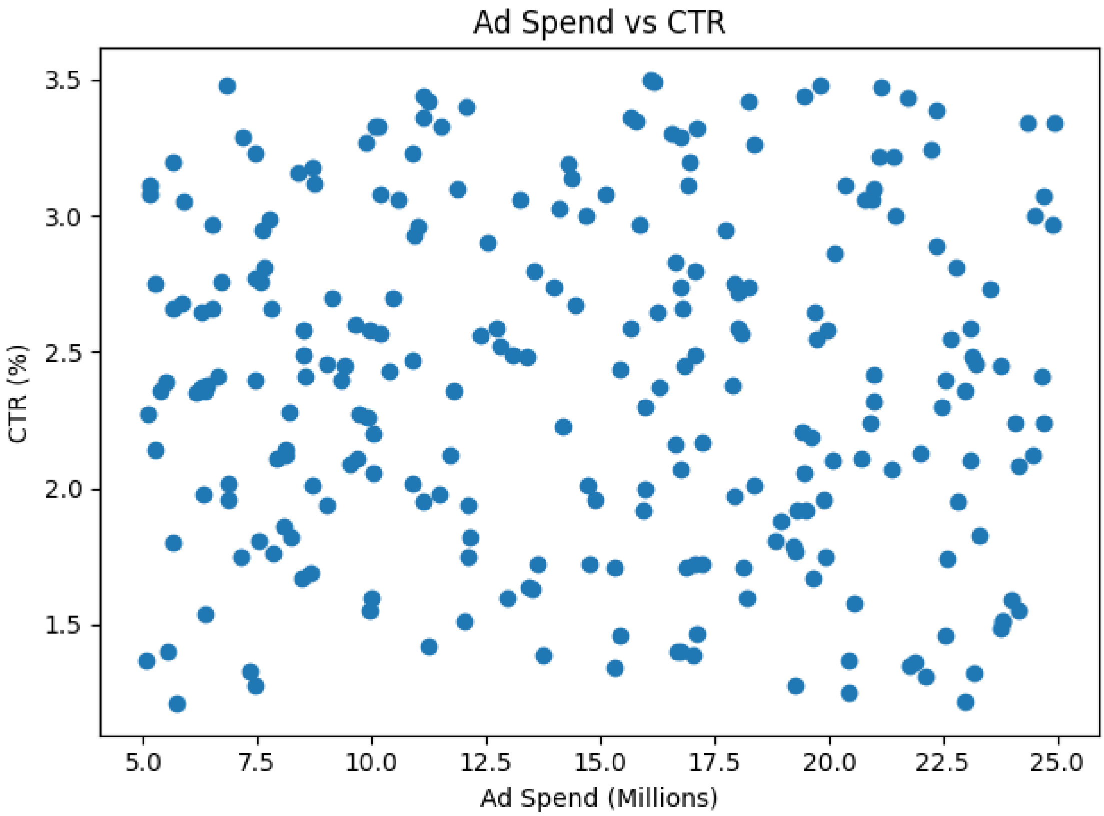


[27]:

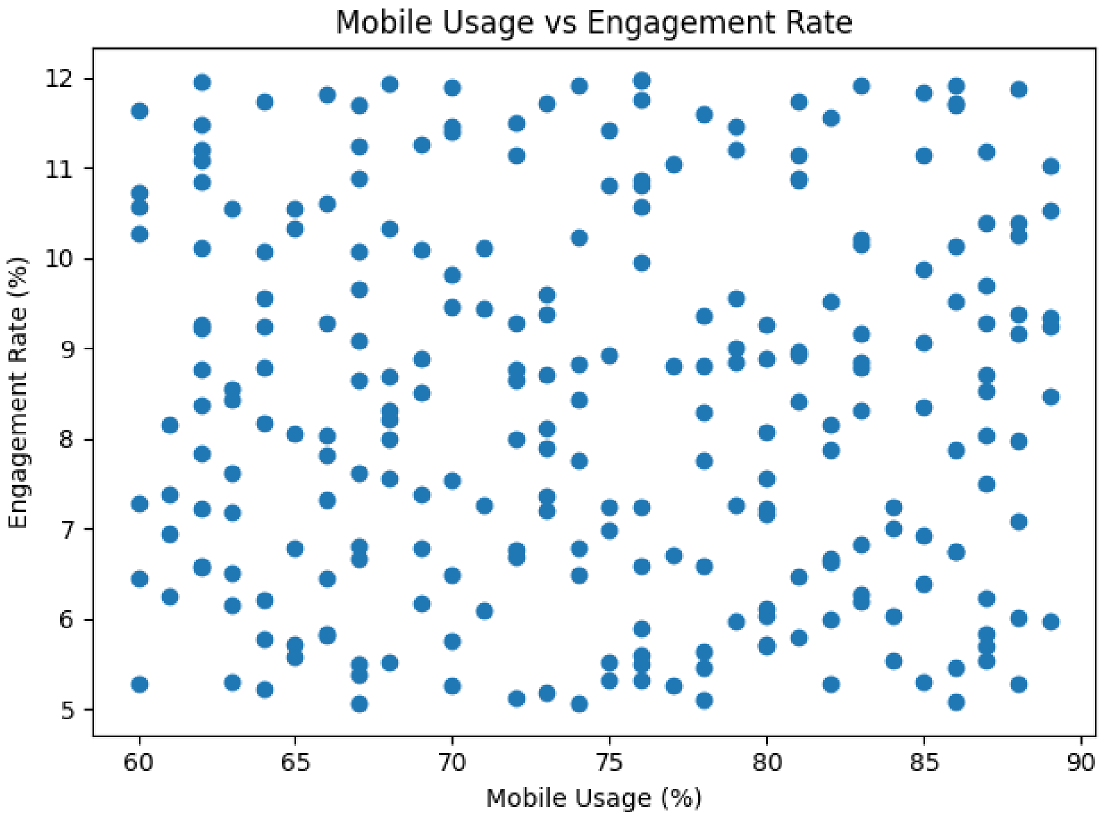
```
# 10. Impressions by State
state_impressions = df.groupby("State")["Ad_Impressions_M"].sum().
 ↪sort_values(ascending=False)

plt.figure()
state_impressions.plot(kind="bar")
plt.title("Total Ad Impressions by State")
plt.xlabel("State")
plt.ylabel("Ad Impressions (Millions)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



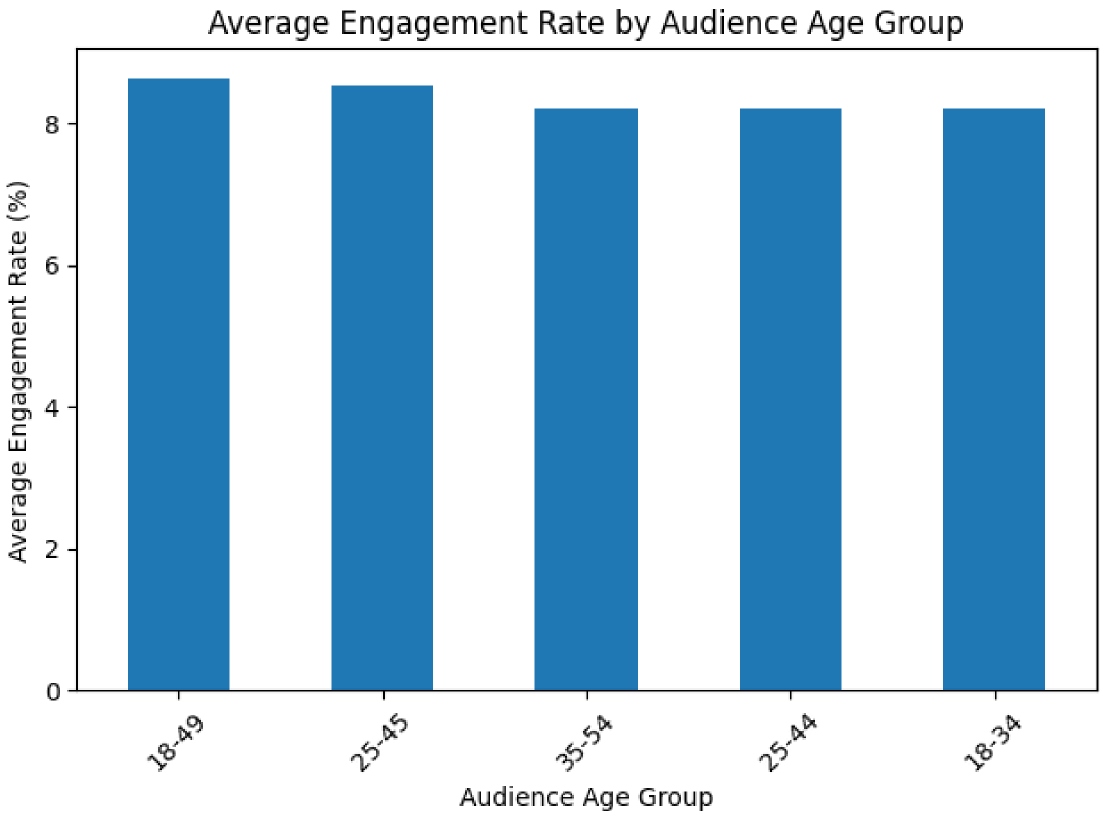
[28]:
```
# 11. Ad Spend vs CTR
plt.figure()
plt.scatter(df["Ad_Spend_M"], df["CTR_%"])
plt.title("Ad Spend vs CTR")
plt.xlabel("Ad Spend (Millions)")
plt.ylabel("CTR (%)")
plt.tight_layout()
plt.show()
```



[29]:
```
# 12. Mobile Usage vs Engagement Rate
plt.figure()
plt.scatter(df["Mobile_Usage_%"], df["Engagement_Rate_%"])
plt.title("Mobile Usage vs Engagement Rate")
plt.xlabel("Mobile Usage (%)")
plt.ylabel("Engagement Rate (%)")
plt.tight_layout()
plt.show()
```



[30]:
```
# 13. Average Engagement by Audience Age Group
age_engagement = df.groupby("Audience_Age_Group")["Engagement_Rate_%"].mean().
 ↪sort_values(ascending=False)

plt.figure()
age_engagement.plot(kind="bar")
plt.title("Average Engagement Rate by Audience Age Group")
plt.xlabel("Audience Age Group")
plt.ylabel("Average Engagement Rate (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

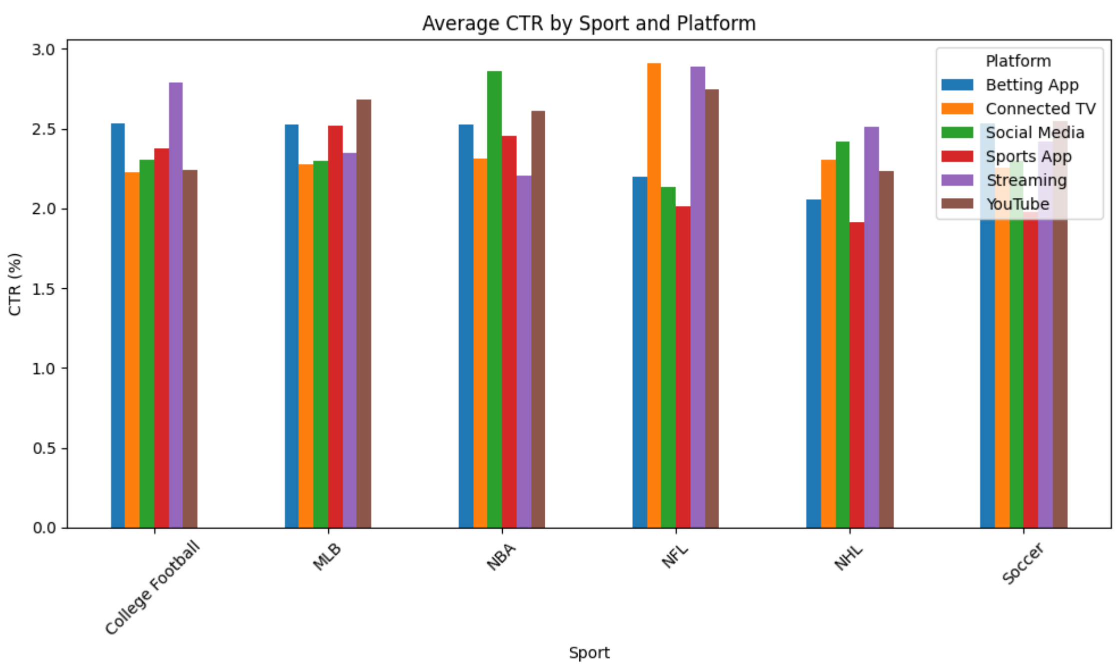


[31]:
```python
# 14. Average CTR by Sport and Platform
sport_platform_ctr = df.pivot_table(
    values="CTR_%",
    index="Sport",
    columns="Platform",
    aggfunc="mean"
)

sport_platform_ctr
```

[31]:
```
Platform          Betting App  Connected TV  Social Media  Sports App  \
Sport
College Football     2.530000      2.224286      2.308000    2.376250
MLB                  2.525556      2.275714      2.295000    2.518571
NBA                  2.527143      2.315833      2.857500    2.451818
NFL                  2.196667      2.912000      2.137500    2.012000
NHL                  2.056667      2.305000      2.416667    1.914000
Soccer               2.530000      2.252857      2.296000    1.978571

Platform          Streaming   YouTube
Sport
```

```
College Football   2.786000  2.240000
MLB                2.345000  2.683333
NBA                2.203333  2.608571
NFL                2.890000  2.743333
NHL                2.515000  2.231250
Soccer             2.416250  2.546250
```

[32]:
```python
# 15. Plot average CTR by sport and platform
sport_platform_ctr.plot(kind="bar", figsize=(10, 6))
plt.title("Average CTR by Sport and Platform")
plt.xlabel("Sport")
plt.ylabel("CTR (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



[33]:
```python
# 16. Top 10 highest impression records
top_impressions = df.sort_values(by="Ad_Impressions_M", ascending=False).
↪head(10)
top_impressions
```

[33]:
```
                Sport      Platform  Game_Moment State Audience_Age_Group  \
116               NBA    Sports App     Pre-Game    OH              18-34
169               NBA    Sports App  2nd Quarter    NY              35-54
6    College Football     Streaming  1st Quarter    FL              25-44
```

```
178                NFL   Betting App     Halftime    MI                25-45
129                NBA  Connected TV     Halftime    MI                18-49
110   College Football  Social Media     Pre-Game    MI                18-49
22                 NFL  Social Media    Post-Game    CA                18-34
175                NBA    Sports App  3rd Quarter    NY                18-34
198                NBA  Social Media  1st Quarter    TX                25-45
237   College Football     Streaming  1st Quarter    NY                35-54

     Mobile_Usage_%  Ad_Impressions_M  CTR_%  Engagement_Rate_%  Ad_Spend_M  \
116              67               599   2.66               6.81        7.79
169              81               596   1.67              10.88        8.48
6                69               595   3.33              10.09       11.53
178              86               594   1.28              11.91        7.46
129              77               592   3.11               5.26       16.93
110              83               592   2.36               9.17       11.79
22               70               590   1.46              11.47       22.55
175              67               589   2.11               6.66        9.69
198              69               588   2.55               6.18       19.74
237              74               585   3.08               6.49       15.12

     Ad_Recall_%
116           17
169           11
6             13
178           11
129            9
110           11
22             9
175           16
198            8
237           14
```

[34]:

```python
# 17. Top 10 highest engagement records
top_engagement = df.sort_values(by="Engagement_Rate_%", ascending=False).
 ↪head(10)
top_engagement
```

[34]:

```
                 Sport      Platform  Game_Moment State Audience_Age_Group  \
66                 NFL       YouTube    Post-Game    OH              25-45
60                 NHL  Social Media  1st Quarter    MI              18-34
204                MLB  Connected TV     Halftime    NY              25-44
48    College Football  Social Media     Halftime    CA              35-54
8                  NHL   Betting App  3rd Quarter    PA              35-54
178                NFL   Betting App     Halftime    MI              25-45
161             Soccer  Social Media  2nd Quarter    OH              18-49
236             Soccer  Connected TV    Post-Game    MI              25-45
136                NHL   Betting App  2nd Quarter    CA              35-54
```

```
163  College Football  Connected TV     Pre-Game     NY              25-45

     Mobile_Usage_%  Ad_Impressions_M  CTR_%  Engagement_Rate_%  Ad_Spend_M  \
66               76               346   3.50              11.98       16.11
60               76               163   1.88              11.97       18.94
204              62               491   2.74              11.96       18.24
48               68               557   3.06              11.93       13.25
8                74               450   1.37              11.91       20.44
178              86               594   1.28              11.91        7.46
161              83               190   1.76              11.91        7.85
236              70               405   2.45              11.89       23.75
136              88               576   2.55              11.88       22.68
163              85               555   1.94              11.84        9.02

     Ad_Recall_%
66            11
60             9
204            7
48             8
8             18
178           11
161           12
236           19
136           14
163           16
```

[35]:
```python
# 18. Save summary
sport_engagement.to_csv("sport_engagement_summary.csv")
platform_ctr.to_csv("platform_ctr_summary.csv")
moment_recall.to_csv("game_moment_recall_summary.csv")
state_impressions.to_csv("state_impressions_summary.csv")
```

[36]:
```python
# 19. correlation table for numeric columns
df[[
    "Mobile_Usage_%",
    "Ad_Impressions_M",
    "CTR_%",
    "Engagement_Rate_%",
    "Ad_Spend_M",
    "Ad_Recall_%"
]].corr()
```

[36]:
```
                   Mobile_Usage_%  Ad_Impressions_M     CTR_%  \
Mobile_Usage_%           1.000000          0.043659 -0.010714
Ad_Impressions_M         0.043659          1.000000 -0.082877
CTR_%                   -0.010714         -0.082877  1.000000
Engagement_Rate_%       -0.010744          0.015801  0.178926
```

```
Ad_Spend_M             -0.074396         -0.024705 -0.052538
Ad_Recall_%            -0.073255          0.018445 -0.001192

                   Engagement_Rate_%  Ad_Spend_M  Ad_Recall_%
Mobile_Usage_%             -0.010744   -0.074396    -0.073255
Ad_Impressions_M            0.015801   -0.024705     0.018445
CTR_%                       0.178926   -0.052538    -0.001192
Engagement_Rate_%           1.000000   -0.060036     0.071287
Ad_Spend_M                 -0.060036    1.000000    -0.042482
Ad_Recall_%                 0.071287   -0.042482     1.000000
```

[37]:

```python
# 20. Final written insights
print("Key Insights:")
print("- This analysis identifies which sports and platforms produce the strongest engagement and ad performance.")
print("- It also shows how ad recall changes by game moment and how impressions vary by state.")
print("- Scatter plots help evaluate whether ad spend and mobile usage are associated with stronger performance metrics.")
```

```
Key Insights:
- This analysis identifies which sports and platforms produce the strongest
engagement and ad performance.
- It also shows how ad recall changes by game moment and how impressions vary by
state.
- Scatter plots help evaluate whether ad spend and mobile usage are associated
with stronger performance metrics.
```